Unit 2 Progress Check: MCQ Part A

1. The variables x, y, and z are integers. Which of the following algorithms can be used to swap the values of the variables so that they are ordered from least to greatest (i.e., $x \leq y \leq z$)?

(A)

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.

(B)

Step 1:	If x is greater than z, swap the values of x and z.
Step 2:	If y is greater than z, swap the values of y and z.

(C)

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.
Step 3:	If x is greater than y, swap the values of x and y.

(D)

Step 1:	If x is greater than y, swap the values of x and y.
Step 2:	If y is greater than z, swap the values of y and z.
Step 3:	If x is greater than z, swap the values of x and z.

2. A company is designing a program to evaluate the strength of a user's password. The program needs to indicate if the password is "weak," "medium," or "strong" based on the following guidelines.

- A password with fewer than 8 characters is classified as "weak."
- A password with 8 or more characters is classified as "medium," where all characters are either numbers or letters.
- A password with 8 or more characters plus at least one special character (like !, @, or #) is classified as "strong."

Which of the following algorithms will correctly classify a password?

(A) Step 1: If the password has fewer than 8 characters, label it as "weak" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password has more than 8 characters, label it as "medium" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "strong."

(B) Step 1: If the password has fewer than 8 characters, label it as "weak" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password contains at least one special character, label it as "strong" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "medium."

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(C) Step 1: If the password has 8 or more characters, label it as "medium" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password has more than 8 characters and contains at least one special character, label it as "strong" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "weak."

(D) Step 1: If the password has 8 or more characters and contains at least one special character, label it as "strong" and skip Steps 2 and 3. Otherwise, go to Step 2.

Step 2: If the password does not contain at least one special character, label it as "medium" and skip Step 3. Otherwise, go to Step 3.

Step 3: Label the password as "weak."

3. Users of a mobile application are categorized based on the average number of hours they use the application each day, as shown in the following table.

Category	Average Hours of Use Per Day
Top User	Greater than 5
Regular User	Between 1 and 5, inclusive
Inactive User	Less than 1

Suppose *n* is the average number of hours a user spends using the application each day. Which of the following algorithms can be used to assign the correct categorization to a user?

(A) If *n* is greater than 5, categorize the user as a top user. Otherwise, if *n* is less than or equal to 5, categorize the user as a regular user. Otherwise, categorize the user as an inactive user.

(B) If *n* is greater than 5, categorize the user as a top user. Otherwise, if *n* is greater than or equal to 1, categorize the user as a regular user. Otherwise, categorize the user as an inactive user.

(C) If *n* is less than 1, categorize the user as an inactive user. Otherwise, if *n* is greater than or equal to 1, categorize the user as a regular user. Otherwise, categorize the user as a top user.

(D) If *n* is greater than or equal to 1, categorize the user as a regular user. Otherwise, if *n* is less than 1, categorize the user as an inactive user. Otherwise, categorize the user as a top user.

4. Consider the following code segment, which is intended to print `true` only when both `r` and `v` are positive numbers. Assume that `r` and `v` have been properly declared and initialized.

```
boolean rPos = r > 0;
boolean vPos = v > 0;
System.out.print(rPos == vPos);
```

The code segment does not always work as intended.

Which of the following values for `r` and `v` will demonstrate that this code segment does **not** work as intended?

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(A)
r = -10, v = 10

(B)
r = -1, v = -4

(C)
r = 5, v = -2

(D)
r = 6, v = 3

5. In the following code segment, a and b are properly declared and initialized boolean variables.

```
boolean result = (a != b);
System.out.println(result);
```

Which of the following best describes the behavior of the code segment?

(A) It prints true if a is true and b is false, and prints false otherwise.
(B) It prints true if a and b have different values, and prints false otherwise.
(C) It prints true if a and b have the same value, and prints false otherwise.
(D) It prints false for all possible values of a and b.

6. Consider the following expression.

$(3 + 4 == 5) != (3 + 4 >= 5)$

What value, if any, does the expression evaluate to?

(A)
true

(B)
false

(C)
5

(D) No value; relational operators cannot be used on arithmetic expressions.

7. In the following code segment, n is a properly declared and initialized int variable.

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```
boolean result = false;
if (n >= 10)
{
  result = true;
}
if (n <= 50)
{
  result = true;
}
System.out.println(result);
```

Which of the following best describes the behavior of the code segment?

(A) It prints true for all possible values of n.
(B) It prints false for all possible values of n.
(C) It prints true if n is between 10 and 50, inclusive, and prints false otherwise.
(D) It prints false if n is between 10 and 50, inclusive, and prints true otherwise.

8. Consider the following code segment.

```
int quant = 20;
int unitPrice = 4;
int ship = 8;
int total;
if (quant > 10)
{
  unitPrice = 3;
}
if (quant > 20)
{
  ship = 0;
}
total = quant * unitPrice + ship;
```

What is the value of total after this code segment is executed?

(A)
20

(B)
60

(C)
68

(D)
88

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9. The following code segment is intended to print the total cost of a stay in a room at a particular hotel. The int variable numNights represents the length of the stay, in nights, and the double variable dailyRate represents the room base cost for each night of the stay. For stays that are longer than five nights, there is a 10% discount applied to the room base cost for all nights during the stay. In addition to the room cost, the hotel charges a $25 resort fee for each night of the stay. The resort fee is not eligible for the 10% discount for stays that are longer than five nights.

```
int numNights =   /* some initial value */ ;
double dailyRate = /* some initial value */ ;

double totalCost = numNights * dailyRate;
/* missing code */

System.out.println(totalCost);
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

(A)

```
if (numNights > 5)
{
  totalCost *= 0.9;
  totalCost += (numNights * 25);
}
```

(B)

```
if (numNights > 5)
{
  totalCost *= 0.9;
}
totalCost += (numNights * 25);
```

(C)

```
totalCost += (numNights * 25);
if (numNights > 5)
{
  totalCost *= 0.9;
}
```

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(D)

```
totalCost += 25;
if (numNights > 5)
{
  totalCost *= 0.9;
}
```

10. Consider the following two code segments. Assume that variables x and y have been declared as int variables and have been assigned integer values.

Code Segment I

```
int result = 0;
if (x > y)
{
  result = x - y;
  System.out.print(result);
}
else if (x < y)
{
  result = y - x;
  System.out.print(result);
}
else
{
  System.out.print(result);
}
```

Code Segment II

```
if (x < y)
{
  System.out.print(y - x);
}
else
{
  System.out.print(x - y);
}
```

Which of the following correctly compares the outputs of the two code segments?

(A) Code segment I and code segment II produce the same output for all values of x and y.
(B) Code segment I and code segment II produce the same output only when x is equal to y.
(C) Code segment I and code segment II produce the same output only when x is not equal to y.
(D) Code segment I and code segment II produce the same output only when x is less than y.

11. Consider the following code segment, which uses properly declared and initialized int variables x and y and the String variable result.

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```
String result = "";
if (x < 5)
{
  if (y > 0)
  {
    result += "a";
  }
  else
  {
    result += "b";
  }
}
else if (x > 10)
{
  if (y < 0)
  {
    result += "c";
  }
  else if (y < 10)
  {
    result += "d";
  }
  result += "e";
}
result += "f";
```

What is the value of result after the code segment is executed if x has the value 15 and y has the value 5?

(A)
adf

(B)
d

(C)
def

(D)
ef

12. In the following code segment, str is a properly declared and initialized String variable.

```
int result = 0;
if (str.length() > 5)
{
  if (str.indexOf("A") < 0)
  {
```

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```
    result = 1;
  }
  else if (str.indexOf("B") < 0)
  {
    result = 2;
  }
}
else if (str.indexOf("A") < 0)
{
  result = 3;
}
```

Which of the following code segments assigns the same value to result as the preceding code segment for all values of str?

(A)

```
int result = 0;
if (str.indexOf("A") < 0)
{
  result = 3;
}
else if (str.indexOf("B") < 0)
{
  result = 2;
}
else if (str.length() > 5);
{
  result = 1;
}
```

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(B)

```
int result = 0;
if (str.indexOf("A") < 0)
{
  if (str.length() > 5)
  {
    result = 1;
  }
  else if (str.indexOf("B") < 0)
  {
    result = 2;
  }
  else
  {
    result = 3;
  }
}
```

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(C)

```
int result = 0;
if (str.indexOf("A") < 0)
{
  if(str.length() > 5)
  {
    result = 1;
  }
  else
  {
    result = 3;
  }
}
else if (str.indexOf("B") < 0)
{
  if (str.length() > 5)
  {
    result = 2;
  }
}
```

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(D)

```
int result = 0;
if (str.indexOf("B") < 0)
{
  if (str.length() > 5)
  {
    result = 2;
  }
}
else if (str.indexOf("A") < 0)
{
  if (str.length() > 5)
  {
    result = 1;
  }
  else
  {
    result = 3;
  }
}
```

13. A certain device requires a passcode to be used. The passcode must meet both of the following requirements.

- The passcode must be 4 digits long.
- The passcode must not start with a zero (0) digit.

Assume that `pass` is a properly declared and initialized `int` variable.

Which of the following expressions will evaluate to `true` if `pass` is an acceptable passcode?

(A)

```
pass >= 1000 || pass < 10000
```

(B)

```
pass > 1000 || pass <= 10000
```

(C)

```
pass >= 1000 && pass < 10000
```

(D)

pass > 1000 && pass < 10000

14. A fitness application categorizes daily step counts into different activity levels. The "Active" level is defined as any step count between 10,000 and 15,000, inclusive. The following code segment is intended to determine if a user's step count falls within the "Active" category.

```
if (steps > 10000 && steps < 15000)
{
  activityLevel = "Active";
}
```

Which of the following best explains the error, if any, in this code segment?

(A) There is no error in the code segment as it correctly identifies "Active" step counts.

(B) The condition should be `if (steps >= 10000 && steps <= 15000)` to properly include 10,000 and 15,000.

(C) The condition should be `if (steps > 10000 || steps < 15000)` to include the correct range of step counts.

(D) The condition should be `if (steps >= 10000 || steps <= 15000)` to include all step counts from 10,000 and above, as well as 15,000 and below.

15. Consider the following code segment.

```
boolean a = true;
boolean b = true;
System.out.print((b || (!a || b)) + " ");
System.out.print(((!b || !a) && a) + " ");
System.out.println(!(a && b) && b);
```

What output is produced when this code segment is executed?

(A)

true true true

(B)

true false true

(C)

true false false

(D)

false true false

16. In the following code segment, `x` is a properly declared and initialized `boolean` variable and `y` is a properly declared and initialized `int` variable.

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```
boolean a = !x && y > 50000;
boolean b = !(x || y <= 50000);
boolean c = !x || y <= 50000;
boolean d = !(x && y <= 50000);

if (a == b)
{
  System.out.println("First");
}
if (a == c)
{
  System.out.println("Second");
}
if (b == c)
{
  System.out.println("Third");
}
if (c == d)
{
  System.out.println("Fourth");
}
```

Which of the following is always printed as a result of executing this code segment?

(A)
First

(B)
Second

(C)
Third

(D)
Fourth

17. An image on a computer is made up of many pixels. Each pixel in the image contains varying amounts of red, green, and blue. A "mainly red" pixel is defined to be one in which the percentage of red is more than 50, and the percentages of green and blue are each less than 10.

Assume that `redPct`, `greenPct`, and `bluePct` are `int` variables that have been properly declared and initialized with values representing the percentages of red, green, and blue, respectively, in a pixel.

Which of the following expressions will evaluate to `true` for a "mainly red" pixel?

(A)
`(redPct <= 50) || (greenPct >= 10) || (bluePct >= 10)`

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(B)

```
(redPct > 50) && (greenPct < 10) || (bluePct < 10)
```

(C)

```
!( (redPct <= 50) || (greenPct >= 10) || (bluePct >= 10) )
```

(D)

```
!(redPct > 50) && !(greenPct > 10) && !(bluePct > 10)
```

18. In the following code segment, str1 and str2 are String objects. The code segment is intended to print true if str1 and str2 are non-null strings that contain the same sequence of characters. The code segment is intended to print false otherwise.

```
boolean result = false;          // line 1
if(str1 != null && str2 != null)  // line 2
{
  result = str1 == str2;         // line 4
}
System.out.println(result);
```

Which of the following best explains the error, if any, in the code segment?

(A) In line 1, result should be initialized to true.
(B) In line 2, str1 and str2 should be compared to null with the equals method.
(C) In line 4, str1 and str2 should be compared to each other with the equals method.
(D) The code segment contains no errors and works as intended.